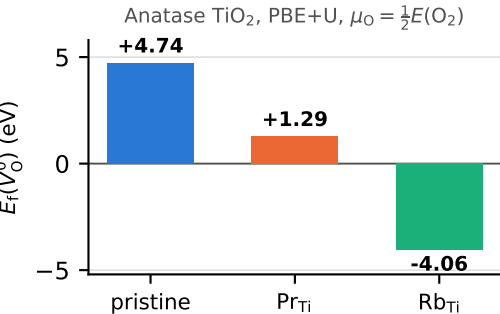




Oxygen vacancies in Pr- and Rb-substituted anatase TiO_2

Both dopants lower the vacancy cost; Pr stays uphill, Rb goes downhill: a 5.36 eV contrast.

The order follows the formal hole count (0, 1, 3 per cell); hole localisation in the Pr cell depends on an O-2p Hubbard term.

Same supercell, pseudopotentials, U and O_2 reference